

Cite this: *Nanoscale*, 2018, **10**, 21320

An artificial intelligence atomic force microscope enabled by machine learning†

Boyuan Huang,^a Zhenghao Li^{a,b} and Jiangyu Li^{ID} *^{a,c}

Artificial intelligence (AI) and machine learning have promised to revolutionize the way we live and work, and one of the particularly promising areas for AI is image analysis. Nevertheless, many current AI applications focus on the post-processing of data, while in both materials sciences and medicine, it is often critical to respond to the data acquired on the fly. Here we demonstrate an artificial intelligence atomic force microscope (AI-AFM) that is capable of not only pattern recognition and feature identification in ferroelectric materials and electrochemical systems, but can also respond to classification *via* adaptive experimentation with additional probing at critical domain walls and grain boundaries, all in real time on the fly without human interference. Key to our success is a highly efficient machine learning strategy based on a support vector machine (SVM) algorithm capable of high fidelity pixel-by-pixel recognition instead of relying on the data from full mapping, making real time classification and control possible during scanning, with which complex electromechanical couplings at the nanoscale in different material systems can be resolved by AI. For AFM experiments that are often tedious, elusive, and heavily rely on human insight for execution and analysis, this is a major disruption in methodology, and we believe that such a strategy empowered by machine learning is applicable to a wide range of instrumentations and broader physical machineries.

Received 20th August 2018,
Accepted 20th October 2018

DOI: 10.1039/c8nr06734a

rsc.li/nanoscale

Introduction

Artificial intelligence (AI) and machine learning have promised to revolutionize the way we live and work. In the ancient game of Go, AI has shed unprecedented new insights that have not been recognized by mankind over several thousand years;¹ in medicine, AI has offered diagnosis that rivals the best human doctors;^{2,3} and in physics as well as materials sciences, AI has enabled accelerated discovery of new substances, compounds, and mechanisms.^{4–8} One of the particularly promising areas for AI is image analysis,^{9,10} where it far outperforms human beings in pattern recognition, capable of discerning subtle features that are elusive to the naked eye. Indeed, AI has been demonstrated to be very effective in analyzing data from both microscopy and spectroscopy studies.^{6,11–14} Nevertheless, many current AI applications in image analysis focus on the post-processing of data,^{12,15–17} while in both materials sciences and medicine, especially in time- and environment-sensitive

circumstance and at elusive points that are not easy to spot, it is often critical to respond to the data acquired on the fly, for example by acquiring additional data in the critical locations of material interfaces or tumors. It is also highly desirable to intervene in real time with manipulative or therapeutic treatments on the spot. Here we develop a strategy toward this vision, by demonstrating an artificial intelligence atomic force microscope (AI-AFM) that is capable of not only pattern recognition and feature identification, but can also respond to classification *via* adaptive experimentation with additional probing at critical locations, all in real time on the fly without human interference. We believe that such a strategy is applicable to a wide range of instrumentations and experiments, embodying the true spirit of the automation of science.^{18,19}

Our work was initially motivated by atomic force microscopy (AFM),²⁰ which is a powerful tool in probing, elucidating, and manipulating materials and structures at the nanoscale. Yet AFM experimentations are very tedious and heavily rely on users' experiences in recognizing usually elusive underlying processes. Very often important yet subtle information is overlooked by the users while conducting experiments, and insights are only realized during the post-data processing afterward, which is often too late – it is virtually impossible to get back to the critical locations again for further probing, such as defects, heterogeneities, and inter-

^aDepartment of Mechanical Engineering, University of Washington, Seattle, WA 98195-2600, USA. E-mail: jili@uw.edu

^bDepartment of Mechanical Engineering, Tsinghua University, Beijing 100084, China

^cShenzhen Key Laboratory of Nanobiomechanics, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen 518055, Guangdong, China

†Electronic supplementary information (ESI) available. See DOI: 10.1039/c8nr06734a

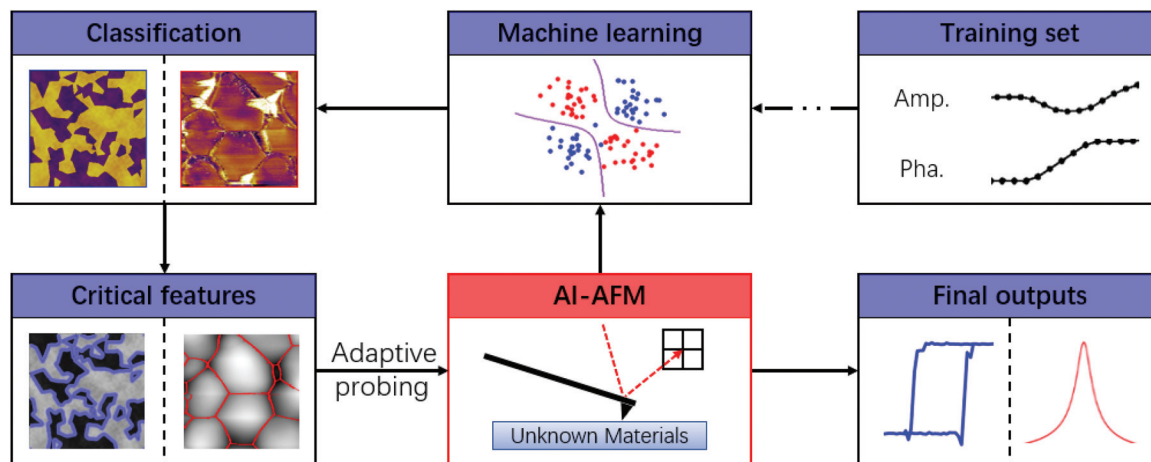


Fig. 1 The concept of AI-AFM that feeds scanning data to machine learning in real time and classifies samples under probing accordingly, with appropriate features identified. Additional experiments are then carried out on the fly near critical spots for additional data and further insight, all without human interference.

faces, where the most interesting physics occur. The sample could be decomposed for example, or the elusive critical points could be lost like a needle in a forest. This is an ideal scenario for the AI-AFM we propose, which is capable of not only recognition and classification, but can also follow up with additional probing in real time upon critical features for further insight, saving all the trouble for human users. This vision is schematically shown in Fig. 1, consisting of an AI-AFM that feeds scanning data to a machine learning algorithm in real time. The algorithm is pre-trained with data for material classification and feature recognition, and based on a particular class of materials recognized by AI, additional features that are relevant to the underlying system will be identified on the fly, such as domain walls (DWs) or grain boundaries (GBs), among others. Through the control algorithm, the probe will get back to the identified critical feature in real time and carry out further experimentation appropriate for the probed system on the fly, yielding additional data for analyzing the underlying physical processes. Key to such a vision is a highly efficient machine learning algorithm capable of high fidelity pixel-by-pixel recognition instead of relying on the data from full mapping, so that real time classification and control are made possible during scanning. We emphasize that the concept of the AI-AFM is fundamentally different from that of the AFM robot,²¹ which also utilizes an intensive imaging process, yet still relies on user intervention through an augmented reality system. In the AI-AFM, all the sophisticated tasks are accomplished automatically in an artificial intelligence manner without user interference, embodying the true spirit of AI in an AFM.

Results and discussion

To demonstrate the concept, we consider dynamic strain-based scanning probe microscopy (ds-SPM) that is widely used

to probe electromechanical coupling at the nanoscale, including piezoresponse force microscopy (PFM)^{22,23} and electrochemical strain microscopy (ESM),^{24–27} both of which excite samples through a charged conductive probe and measure the corresponding local deformation. The electromechanical coupling is ubiquitous in natural materials, synthetic devices, and biological systems, such as ferroelectric materials,^{23,28} lithium ion batteries,^{25–27} and voltage gated ion channels,²⁹ underpinning a wide range of functionalities in information processing, energy conversion, and biological processes. Despite their vast different microscopy mechanisms, these electromechanical couplings often exhibit themselves in ds-SPM as apparent piezoresponse,³⁰ and it is quite challenging for users to discern their dominating microscopic origin. A couple of examples are shown in Fig. 2, wherein the amplitude and phase mappings of dynamic strain for a typical ferroelectric lead zirconate titanate (PZT) and electrochemical LiVO_3 (LVO) are compared. While they have quite different microscopy mechanisms, the mappings closely resemble each other except for some subtle difference: the 180° phase reversal at the interface with a much reduced amplitude as observed in PZT is a signature of the ferroelectric domain wall, which is not present in LVO. The question is whether we can train a machine learning algorithm that first differentiates ferroelectric domains from non-ferroelectric mappings, and then responds with additional probing necessary at critical locations for further analysis, for example identify DWs in ferroelectrics and GBs in electrochemical materials, after which detailed probing relevant to the particular system can be carried out across these important material interfaces.

180° domains are commonly present in ferroelectric materials to minimize their free energy,³¹ and they usually exhibit much reduced piezoresponse on DWs with 180° phase contrast, as revealed in Fig. 2(c). On the other hand, the amplitude and phase behaviors of non-ferroelectric solids, such as electrochemical materials, are usually not well defined as

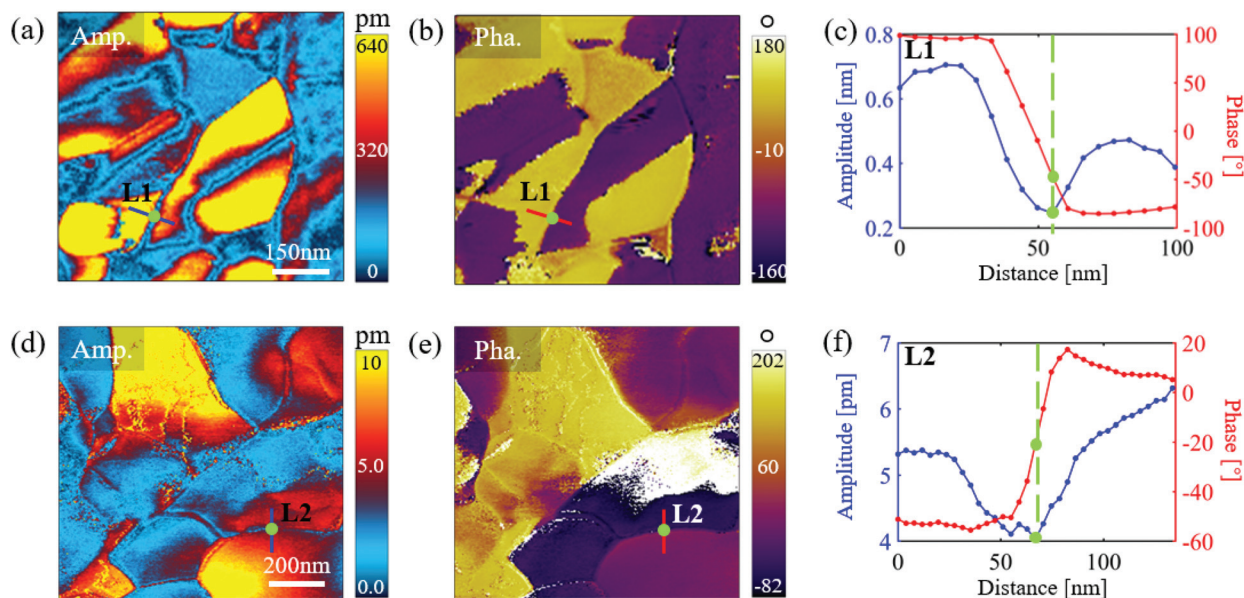


Fig. 2 Comparing the amplitude and phase mappings of dynamic strain measured by ds-SPM for (a–c) ferroelectric PZT and (d–f) electrochemical LiVO_3 , along with their respective line profiles.

revealed in Fig. 2(f), wherein the phase contrast is smaller than 180° . Based on these observations, we employ a support vector machine (SVM) algorithm³² to develop a physics-based classifier that is capable of extracting ferroelectric DWs pixel by pixel from the inputted PFM mappings, thereby helping in distinguishing ferroelectric materials from electrochemical ones, for which a different algorithm is introduced to extract GBs from AFM topography mappings, as detailed in Fig. S1 in the ESI†. Note that while a popular convolutional neural network (CNN) has achieved remarkable success in the field of image recognition,¹⁰ it can only categorize a whole map but is not capable of delineating the exact DWs or GBs of interest. Fully convolutional networks derived from CNN are capable of identifying lattice atoms in raw scanning transmission electron microscopy (STEM) data^{12,33} and should be applicable to our problem, but it requires extensive GPUs to facilitate the training process as well as a large amount of training data with DWs or GBs accurately labelled at the level of pixel. Such sophistications are unnecessary for our particular application, since SVM-based AI algorithm needs only a small dataset that can be trained in less than 10 s on an ordinary PC, making it widely accessible. More importantly, this SVM-based algorithm is much more efficient than CNN, critical for classification and control on the fly. On the other hand, we also note that due to subtleties and complexities often exhibited in ferroelectric and electrochemical materials, careful classification based on machine learning beyond simple rule-based analysis is necessary for our AI-AFM system, and we will come back to this point later with illustrating examples.

SVM is one of the most widely used machine learning algorithms in industry and academia,³⁴ as detailed in the ESI† with its applications ranging from handwritten digit recognition for postal automation services,³⁵ E-mail spam filtering,³⁶ and

accelerating discovery of new piezoelectric materials.⁴ SVM can be easily trained with a set of labeled examples, each of which consists of a fixed number of features ($x_1, x_2, \dots, x_n$) and a y label illustrating whether it belongs to one of the two categories ($y = 1$ or 0). As schematically shown in Fig. 3(a), we first prepare a training dataset for our SVM model, wherein the amplitude and phase variations across a morphology interface are used as indicators to classify whether the interface is a ferroelectric DW or not. For each pixel P_0 (marked as green stars) on PFM maps of ferroelectric LiNbO_3 in Fig. 3(a), 6 adjacent pixels (marked as white dots) are picked from a line centered at P_0 and parallel to its phase gradient. The distance from these 6 pixels to P_0 are fixed and their respective amplitude and phase, 14 features in total, are sufficient to represent the local variance across P_0 . Such 14 features with a label of the pixel (DW or not) are then fed into the SVM model. Since each map contains 256×256 pixels and thus generates almost the same amount of training data (with the exception on the map border), it turns out that only 5 pairs of ds-SPM maps are sufficient to train the SVM model, making it highly efficient compared to CNN that has to use a whole map as one training example. More details about our training set are presented in the ESI† along with the artificially simulated dataset shown in Fig. S2.† SVM first represents these training data as points in a hyperspace, the dimensionality of which depends on the number of features n . It then tries to find a hyperplane $\mathbf{w}^T \cdot \mathbf{x} + b = 0$ to separate these points into two categories, as schematically shown in Fig. 3(a), where $\mathbf{w}$ is the feature weight vector and b is the bias term.³² The points that are most close to the hyperplane from both sides are called the support vectors, as marked in Fig. 3(a), and the larger the distance from these two points to the hyperplane (so-called functional margin), the better the performance of the trained classifier model. Finally,

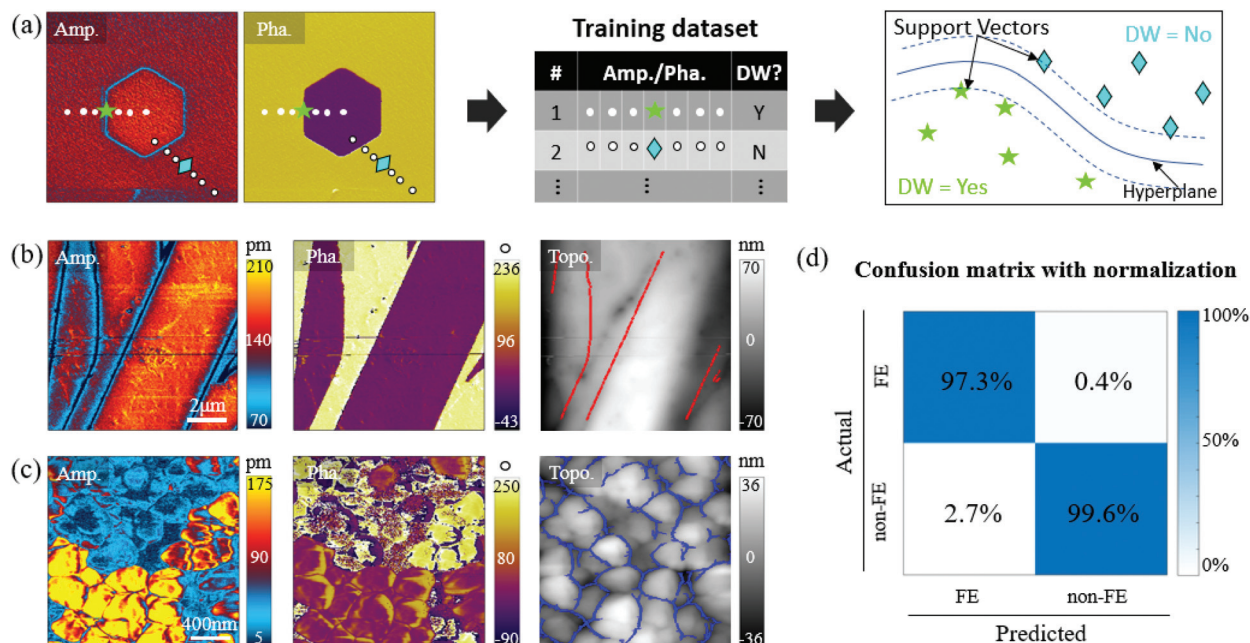


Fig. 3 Illustration and performance of the SVM algorithm for the AI-AFM; (a) schematics of training dataset using PFM mappings of LiNbO_3 and the corresponding classification of ferroelectric DWs; (b) PFM mapping of ferroelectric diisopropylammonium bromide, with 180° DWs identified and overlaid on topography; (c) ESM mapping of electrochemical LiV_3O_8 , with GBs identified and overlaid on topography; (d) normalized confusion matrix of classification performance on 7174 ds-SPM maps of various materials.

testing data from new images that are denoised by a local median filtering with a 3-by-3-pixel window can be mapped into the same space, and then classified according to which side of the hyperplane they fall into, revealing whether it is on a DW or not. Note that the pixel-by-pixel recognition of SVM makes it possible to adaptively adjust experimental parameters during scanning, while for CNN that relies on full mapping for recognition, such real time adjustment is not possible. This process is repeated for all pixels except for those on the border of maps, creating a binary mask with DWs marked as true. Since DWs are continuous lines on the map, the length of the longest line on the binary mask is used to judge whether there are ferroelectric DWs or not. One positive example is shown in Fig. 3(b), wherein 180° DWs in the molecular crystal of diisopropylammonium bromide have been correctly identified by SVM, as marked by the red lines and overlaid on topography. The negative example in Fig. 3(c), on the other hand, shows ESM maps of a LiV_3O_8 sample for which GBs are identified instead. In fact, we have surveyed 7174 ds-SPM mappings accumulated in our lab in the past 10 years, and the normalized confusion matrix in Fig. 3(d) shows that 97.3% of 475 maps predicted having 180° DWs are correctly classified, while 99.6% of 6699 maps predicted having no 180° DWs are properly identified as well. Additional examples are shown in Fig. S3.† These results confirmed that the SVM-based AI algorithm is capable of classification and feature identification of ferroelectric materials with 180° DWs. On the other hand, the popular Canny edge detector often fails, as detailed in Fig. S4.† This is because phase contrast at ferroelectric DWs in real materials often deviates from idealized 180° , and it is

often interfered by topography features as well. Since edge detectors are highly sensitive to the gradient of a map, slight phase distortion due to scanning disturbance, impurities, topography variation, or artificial pattern may cause false identification, as made evident in Fig. S4,† while our machine learning algorithm does not suffer from such problems. This demonstrates the need for machine learning-based AI instead of simple rule-based analysis in classification and identification.

Our ultimate goal is to demonstrate an AI-AFM platform that integrates the SVM algorithm with AFM control that is capable of not only classifying ferroelectric materials with 180° DWs in real time, but also adopt adaptive experimentation on the fly to probe the characteristics and mechanisms of apparent piezoresponse in detail at the critical material interface, *i.e.* DWs in ferroelectrics and GBs in electrochemical materials. Such additional probing is necessary to confirm the classification without ambiguity, a common practice by human users. To this end, two blind experiments have been conducted on ferroelectric PMN-Pt single crystals and electrochemical ceria ceramics, as detailed in the ESI,† both of which are not known to the AI-AFM in advance, and the screenshots of the complete experimental processes are recorded. As shown in Movie S1,† PMN-Pt was determined to be ferroelectric having 180° DWs during scanning, and representative amplitude, phase, and topography (overlaid with identified DWs) mappings are shown in Fig. 4(a and b). After such preliminary classification a “ferroelectric routine” was triggered, with the scanning probe zoomed in on an identified DW and carrying out switching spectroscopy PFM (SS-PFM) experiments³⁷ on a line of

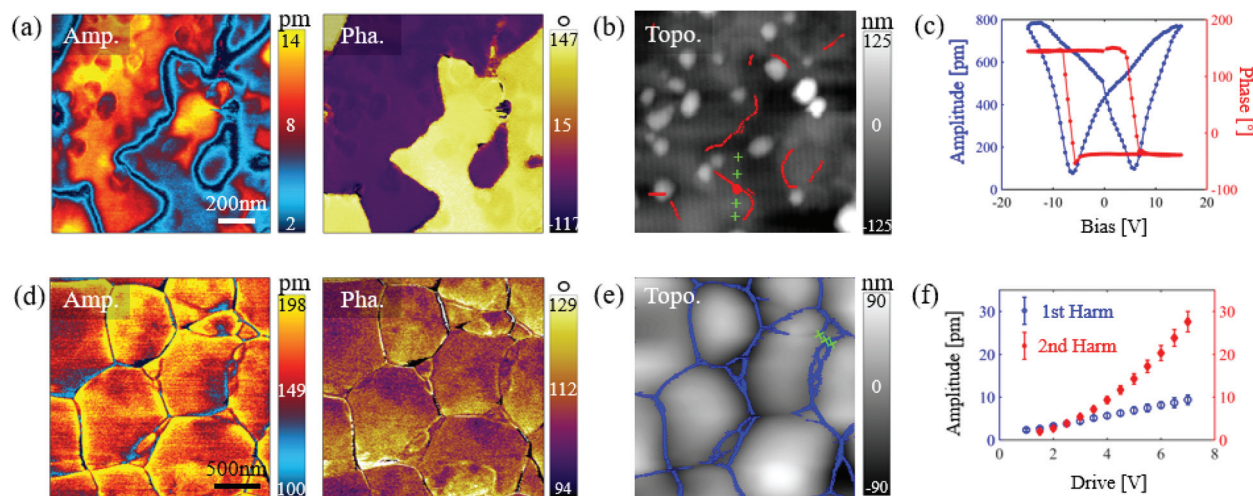


Fig. 4 Demonstration of AI-AFM for two “unknown” samples that are determined to be ferroelectric (a–c) and electrochemical (d–f); (a, d) mappings of amplitude and phase; (b, e) DWs and GBs recognized; and (c, f) ferroelectric hysteresis and butterfly loops on DWs and comparison of first and second harmonic piezoresponse on GBs, all measured on the fly.

points across the DW, yielding hysteresis and butterfly loops as shown in Fig. 4(c), characteristic of ferroelectric materials, and thus confirming the ferroelectric classification without ambiguity. When ceria was tested, as shown in Movie S2,[†] the AI-AFM found no 180° DWs from its amplitude and phase mappings as shown in Fig. 4(d), and thus a “non-ferroelectric routine” was triggered to identify GBs overlaid on topography as shown in Fig. 4(e), after which the scanning probe was zoomed in on an identified GB and measurements of first and second harmonic piezoresponses^{38,39} across GBs were carried out. As seen in Fig. 4(f), second harmonic piezoresponse dominates the first harmonic one in ceria, characteristic of electrochemical materials, and thus confirms its non-ferroelectric nature without ambiguity. As is clear from the movies, both experiments have been conducted in an artificial intelligence manner without human users’ interference, demonstrating the capability of our AI-AFM system. Note that the probed sample could be neither ferroelectric nor electrochemical, which can be revealed by the characteristics of first and second harmonic piezoresponse across grain boundaries.

What we demonstrate here is a simple yet powerful prototype artificial intelligence AFM that is trained to carry out complicated scientific experiments from beginning to end, all on its own, and it is just a matter of time to incorporate more profound physical processes and a more sophisticated deep learning algorithm to expand its power. Indeed, our current implementation is based on the interfacial profile of vertical piezoresponse, which could be complemented by additional lateral probing for possible in-plane polarization, as well as pre-probing lithography for single-domain ferroelectrics. Classifying atypical polar materials such as surface polar phases,⁴⁰ mixed polar/nonpolar-phases,⁴¹ and highly fragmental, curved and fine-scale domains in relaxors⁴² could impose additional challenges, but there is no fundamental difficulty. We also ignored a possible change of surface conditions and

wearing of probe during scanning, which could be addressed by auto-tuning the probe parameters with the help of an advanced convolutional neural network. We acknowledge that the DWs and GBs recognized by the current AI implementation are not really elusive *per se*, but the strategy points toward a viable direction toward more subtle features that are easy to be overlooked. We expect that a similar strategy can also be developed for a wide range of scientific instruments from a transmission electron microscope to an X-ray diffractometer, as well as a broader physical machineries and systems that heavily rely on human experience to operate at the moment. It is also conceivable that an eco-system can emerge from such vision that all the AI-AFMs are interconnected to share and strengthen the training data, machine learning algorithm, and control, so that user experiences and know-how are no longer limited in a particular lab, but readily spread over the network, and we have made our algorithms publicly accessible to facilitate such movement.⁴³ More importantly, a general intelligent machine may evolve from such an eco-system, which is capable of all round AFM experiments and analysis on its own, revolutionizing the way we do AFM experiments.

Methods

Extraction of grain boundaries (GBs)

As shown in the upright inset of Fig. S1(a),[†] the first-order derivative of the height map is usually not continuous at GBs, leading to overwhelming second-order derivative values there compared to those of the nearby area. By setting a threshold value at 90% of the maximum second-order derivative of the map, a binary mask of GBs can be produced as shown in Fig. S1(d).[†] This method works well in general, but may run into difficulty in identifying the exact location of GBs when

there is large topography variation, which calls for more advanced algorithms.

Training data

Considering that the histograms of phase maps are always concentrated around several specific angles, delivering very sparse information, we randomly change the phase offset when picking the 14 features to build a robust model that can work for other realistic cases with various phase offsets. We also intentionally vary the scaling of amplitude features and add white noises to them for the same reason. Although real maps with manually highlighted DWs can be utilized as the training dataset, we also succeed in training a model based on simulated maps without tedious labelling work. Specifically, a binary mask of random DWs is first generated and then rendered to mimic the pattern of real maps with respect to morphology, as shown in Fig. S2.†

Implementation of support vector machine (SVM)

Functions “fitsvm” and “predict” in MATLAB are employed to train SVM models and classify new examples, separately. The Gaussian kernel is used to implicitly map the input dataset into high-dimensional feature spaces,⁴⁴ enabling an efficient non-linear classification. Since amplitude and phase values are two sets of independent features, we use them separately to train two different SVM models – Amp model and Pha model. When doing prediction, the algorithm will assign a pixel as a point of DWs if and only if both Amp and Pha models determine it to be a DW based on input features, as shown in Fig. S3.†

Canny edge detector

It is a technique to extract edges from different vision objects in images.⁴⁵ In this work, we used the edge function in MATLAB with the Canny method and default setting.

First and second harmonics

The first and second harmonic PFM responses originate from piezoresponse and other electromechanical mechanisms.³⁸ For each measured point, a set of AC voltages from 1.5 to 7.5 V were applied with an increment of 0.5 V. At each voltage step, the sample is excited around f_0 first and then around $f_0/2$, thereby generating two sets of tuning data around f_0 . The corresponding first and second harmonic amplitude can be extracted by fitting the raw data with the SHO model.

AI-AFM system

The AI-AFM experimentation is performed with a commercial Asylum Research MFP-3D AFM. Nanosensor PPP-EFM conductive probes were used for all data shown in Fig. 4. The system first conducts a DART PFM mapping to survey possible DWs with the pretrained SVM model. If DWs are found, the probe will move to locate in the middle of the longest DW identified. Then, the system will zoom in on this area with a scan size half of previous one to double-check those DWs. Finally, the middle point on the DW as well as other 4 points across it will

be marked so that a set of SS-PFM tests can be completed on each of these points. If no DWs are found, the material will be further assessed for its apparent piezoresponse mechanism. The system will highlight GBs in a height map by using the method discussed in Fig. S1† and then zoom in on a specific GB to finish a few first and second harmonic comparison experiments. The decision-making process of location here is similar to that of DWs. The corresponding video can be seen in Movies S1 and S2.† AI is implemented with the MATLAB code and can automatically send commands to the AFM after analyzing the scanning data on the fly.

Availability of data and codes

All the data and algorithms of the AI-AFM used in this paper can be requested from authors.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

Support from the National Key Research and Development Program of China (2016YFA0201001), the National Natural Science Foundation of China (11627801 and 11472236), and the National Science Foundation (CBET 1435968) is acknowledged.

References

- 1 D. Silver, J. Schrittwieser, K. Simonyan, I. Antonoglou, A. Huang, A. Guez, T. Hubert, L. Baker, M. Lai and A. Bolton, *Nature*, 2017, **550**, 354.
- 2 A. Esteva, B. Kuprel, R. A. Novoa, J. Ko, S. M. Swetter, H. M. Blau and S. Thrun, *Nature*, 2017, **542**, 115.
- 3 D. S. Kermany, M. Goldbaum, W. Cai, C. C. S. Valentim, H. Liang, S. L. Baxter, A. McKeown, G. Yang, X. Wu, F. Yan, J. Dong, M. K. Prasadha, J. Pei, M. Ting, J. Zhu, C. Li, S. Hewett, J. Dong, I. Ziyar, A. Shi, R. Zhang, L. Zheng, R. Hou, W. Shi, X. Fu, Y. Duan, V. A. N. Huu, C. Wen, E. D. Zhang, C. L. Zhang, O. Li, X. Wang, M. A. Singer, X. Sun, J. Xu, A. Tafreshi, M. A. Lewis, H. Xia and K. Zhang, *Cell*, 2018, **172**, 1122–1131.
- 4 Y. Ruihao, L. Zhen, P. V. Balachandran, X. Deqing, Z. Yumei, D. Xiangdong, S. Jun, X. Dezhen and L. Turab, *Adv. Mater.*, 2018, **30**, 1702884.
- 5 S. V. Kalinin, B. G. Sumpter and R. K. Archibald, *Nat. Mater.*, 2015, **14**, 973–980.
- 6 L. Li, Y. Yang, D. Zhang, Z.-G. Ye, S. Jesse, S. V. Kalinin and R. K. Vasudevan, *Sci. Adv.*, 2018, **4**, eaap8672.

- 7 A. P. Bartók, S. De, C. Poelking, N. Bernstein, J. R. Kermode, G. Csányi and M. Ceriotti, *Sci. Adv.*, 2017, **3**, e1701816.
- 8 E. D. Cubuk, R. Ivancic, S. S. Schoenholz, D. J. Strickland, A. Basu, Z. S. Davidson, J. Fontaine, J. L. Hor, Y. R. Huang and Y. Jiang, *Science*, 2017, **358**, 1033–1037.
- 9 Y. LeCun, Y. Bengio and G. Hinton, *Nature*, 2015, **521**, 436.
- 10 A. Krizhevsky, I. Sutskever and G. E. Hinton, *Adv. Neural Inform. Process. Syst.*, 2012, 1097–1105.
- 11 A. Belianinov, R. Vasudevan, E. Strelcov, A. Ievlev, C. Steed, S. M. Yang, A. Tselev, S. Jesse, M. Biegalski and G. Shipman, *Adv. Struct. Chem. Imaging*, 2015, **1**, 11.
- 12 M. Ziatdinov, O. Dyck, A. Maksov, X. Li, X. Sang, K. Xiao, R. R. Unocic, R. Vasudevan, S. Jesse and S. V. Kalinin, *ACS Nano*, 2017, **11**, 12742–12752.
- 13 A. V. Ievlev, A. Belianinov, S. Jesse, D. P. Allison, M. J. Doktycz, S. T. Retterer, S. V. Kalinin and O. S. Ovchinnikova, *Sci. Rep.*, 2017, **7**, 1–7.
- 14 E. Liotti, C. Arteta, A. Zisserman, A. Lui, V. Lempitsky and P. S. Grant, *Sci. Adv.*, 2018, **4**, eaar4004.
- 15 N. Jean, M. Burke, M. Xie, W. M. Davis, D. B. Lobell and S. Ermon, *Science*, 2016, **353**, 790.
- 16 B. R. Jany, A. Janas and F. Krok, *Nano Lett.*, 2017, **17**, 6520–6525.
- 17 Q. Li, C. T. Nelson, S.-L. Hsu, A. R. Damodaran, L.-L. Li, A. K. Yadav, M. McCarter, L. W. Martin, R. Ramesh and S. V. Kalinin, *Nat. Commun.*, 2017, **8**, 1468.
- 18 R. D. King, J. Rowland, S. G. Oliver, M. Young, W. Aubrey, E. Byrne, M. Liakata, M. Markham, P. Pir, L. N. Soldatova, A. Sparkes, K. E. Whelan and A. Clare, *Science*, 2009, **324**, 85–89.
- 19 G. Carleo and M. Troyer, *Science*, 2017, **355**, 602–606.
- 20 G. Binnig, C. F. Quate and C. Gerber, *Phys. Rev. Lett.*, 1986, **56**, 930.
- 21 B. Song, R. Yang, N. Xi, K. C. Patterson, C. Qu and K. W. Lai, *J. Lab. Autom.*, 2012, **17**, 425.
- 22 D. A. Bonnell, S. V. Kalinin, A. L. Kholkin and A. Gruverman, *MRS Bull.*, 2009, **34**, 648–657.
- 23 J. Li, J.-F. Li, Q. Yu, Q. N. Chen and S. Xie, *J. Mater.*, 2015, **1**, 3–21.
- 24 A. N. Morozovska, E. A. Eliseev, N. Balke and S. V. Kalinin, *J. Appl. Phys.*, 2010, **108**, 53712.
- 25 J. Zhu, L. Lu and K. Zeng, *ACS Nano*, 2013, **7**, 1666–1675.
- 26 S. Y. Luchkin, K. Romanyuk, M. Ivanov and A. L. Kholkin, *J. Appl. Phys.*, 2015, **118**, 72016.
- 27 Q. Nataly Chen, Y. Liu, Y. Liu, S. Xie, G. Cao and J. Li, *Appl. Phys. Lett.*, 2012, **101**, 63901.
- 28 S. V. Kalinin and A. Gruverman, *Scanning probe microscopy: electrical and electromechanical phenomena at the nanoscale*, Springer Science & Business Media, 2007, vol. 1.
- 29 C. M. Armstrong and B. Hille, *Neuron*, 1998, **20**, 371–380.
- 30 R. K. Vasudevan, N. Balke, P. Maksymovych, S. Jesse and S. V. Kalinin, *Appl. Phys. Rev.*, 2017, **4**, 021302.
- 31 J. Y. Li, R. C. Rogan, E. Üstündag and K. Bhattacharya, *Nat. Mater.*, 2005, **4**, 776.
- 32 M. A. Hearst, S. T. Dumais, E. Osuna, J. Platt and B. Scholkopf, *IEEE Intell. Syst. Appl.*, 1998, **13**, 18–28.
- 33 J. Long, E. Shelhamer and T. Darrell, in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2015, pp. 3431–3440.
- 34 I. Steinwart and A. Christmann, *Support vector machines*, Springer Science & Business Media, 2008.
- 35 D. Decoste and B. Schölkopf, *Mach. Learn.*, 2002, **46**, 161–190.
- 36 H. Drucker, D. Wu and V. N. Vapnik, *IEEE Trans. Neural Netw.*, 1999, **10**, 1048–1054.
- 37 S. Jesse, A. P. Baddorf and S. V. Kalinin, *Appl. Phys. Lett.*, 2006, **88**, 1–4.
- 38 Q. Nataly Chen, Y. Ou, F. Ma, J. Li, Q. N. Chen, Y. Ou, F. Ma, J. Li, Q. N. Chen, Y. Ou, F. Ma and J. Li, *Appl. Phys. Lett.*, 2014, **242907**, 1–5.
- 39 J. Yu, E. N. Esfahani, Q. Zhu, D. Shan, T. Jia, S. Xie and J. Li, *J. Appl. Phys.*, 2018, **123**, 155104.
- 40 S. M. Yang, A. N. Morozovska, R. Kumar, E. A. Eliseev, Y. Cao, L. Mazet, N. Balke, S. Jesse, R. K. Vasudevan and C. Dubourdieu, *Nat. Phys.*, 2017, **13**, 812.
- 41 B. Huang, G. Kong, E. N. Esfahani, S. Chen, Q. Li, J. Yu, N. Xu, Y. Zhang, S. Xie and H. Wen, *NPJ Quantum Mater.*, 2018, **3**, 30.
- 42 Q. Li, Y. Liu, R. L. Withers, Y. Wan, Z. Li and Z. Xu, *J. Appl. Phys.*, 2012, **112**, 52006.
- 43 <https://github.com/Boyuan-H/AI-AFM>.
- 44 A. Géron, *Hands-on machine learning with Scikit-Learn and TensorFlow: concepts, tools, and techniques to build intelligent systems*, O'Reilly Media, Inc., 2017.
- 45 J. Canny, *Readings Comput. Vis.*, 1987, pp. 184–203.